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As Covid makes clear, statistics education is a must

Christine Franklin introduces a new framework for statistics education, contending that statistical reasoning skills should “no longer be optional” for school-age students

In a February 2020 interview with *Significance*, I discussed the necessity of teaching statistics and data science at the school level.¹ Who could have imagined that soon after that publication, the world would go into a shutdown to deal with the Covid-19 pandemic? We are still in a state of disruption due to coronavirus, but if the pandemic has taught us anything, it is that an understanding of, and ability to reason with, statistical information is more important than it has perhaps ever been.

The international community is being called upon to process and make sense of data – from screening test results and risk analysis regarding vaccines, to statistical models predicting numbers of cases, deaths, hospitalisations, and time to herd immunity. Data is also being used to recommend best practices to combat the spread of the virus. Meanwhile, data is called on to address other global issues such as climate change, the economy, elections, and social justice. Surely, it should no longer be optional for our school-age students to develop statistical reasoning skills. It should be mandatory that all students leave secondary school prepared to live and work in a data-driven society. Our students must be able to evaluate all the statistical information encountered on a daily basis: asking questions about the design of the study and interrogating the data, asking if the data collected is useful for answering the statistical investigative question and if the analysis employed is appropriate, and evaluating whether the conclusions made are reasonable.

Since the *Significance* interview was published, the American Statistical Association (ASA) and the National Council for the Teachers of Mathematics have supported, endorsed, and released, in November 2020, the updated *Pre-K–12 Guidelines for Assessment and Instruction in Statistics*

Education II: A Framework for Statistics and Data Science Education, otherwise known as GAISE II (bit.ly/asaGAISE). The original version of GAISE, released in 2005 and with slight revisions in 2007, provided a research-based framework that drew on decades of work from many in statistics education internationally (bit.ly/3jGyq9s). It was a seminal and visionary document that advocated the necessity of data and statistical literacy beginning in the earliest school grades. GAISE I introduced a two-dimensional framework that outlined the statistical problem-solving process (formulate statistical investigative questions, collect/consider data, analyse data, interpret results) across three development levels (A, B and C). Students progress from level A where they collect data, often about themselves, analysing the data descriptively to answer their statistical investigative questions. As they move to level B, students gain more tools to collect/consider data on a larger population and to analyse the data as they attempt to answer their investigative questions. Level B students also begin developing informal inferential skills. At level C, students further develop statistical concepts and use more tools, becoming comfortable with both descriptive and inferential analysis of data to answer statistical investigative questions. The GAISE framework promotes engaging students in the four-step statistical problem-solving process as the big umbrella for approaching all statistical problems and data analysis.

GAISE II builds on these foundations by acknowledging the many changes in computing since 2005 and, most importantly, the rapidly changing types of data we encounter. The revised GAISE incorporates relevant research conducted since GAISE I and makes stronger connections to statistical thinking in the science classroom and to computational thinking, drawing attention

to data science topics such as working with non-traditional data types and classification techniques. (Two examples from GAISE II that highlight these changes can be found in a post at the Ask Good Questions blog; bit.ly/3agiUy9). The statistical problem-solving process and the spirit of GAISE I remain the same, but GAISE II provides enhanced recommendations and examples that illustrate multivariable thinking beginning at an early age, the role of questioning throughout the statistical problem-solving process, the importance of effective communication, the role of probability with statistical reasoning, and the role of assessment.

The vision of GAISE II is that every individual can, with confidence, grapple with fast-changing data and statistical information, be it global or personal, and that they understand the importance of being a healthy sceptic by asking good questions. The desire is that GAISE II will cultivate efforts in championing the ultimate goal: statistical literacy for all. ■

Note

GAISE II is available for free download at bit.ly/asaGAISE. The printed book is available for purchase at Amazon. An introduction to GAISE II is provided in the *Harvard Data Science Review*,² and articles specific to levels A, B and C will appear in forthcoming issues of *HDSR*.

Disclosure statement

The author declares no conflicts of interest.

References

1. Tarran, B. (2020) “Statistical literacy for all!” *Significance*, **17**(1), 42–43.
2. Franklin, C. and Bargagliotti, A. (2020) Introducing GAISE II: A guideline for precollege statistics and data science education. *Harvard Data Science Review*, **2**(4). doi: 10.1162/99608f92.246107bb.